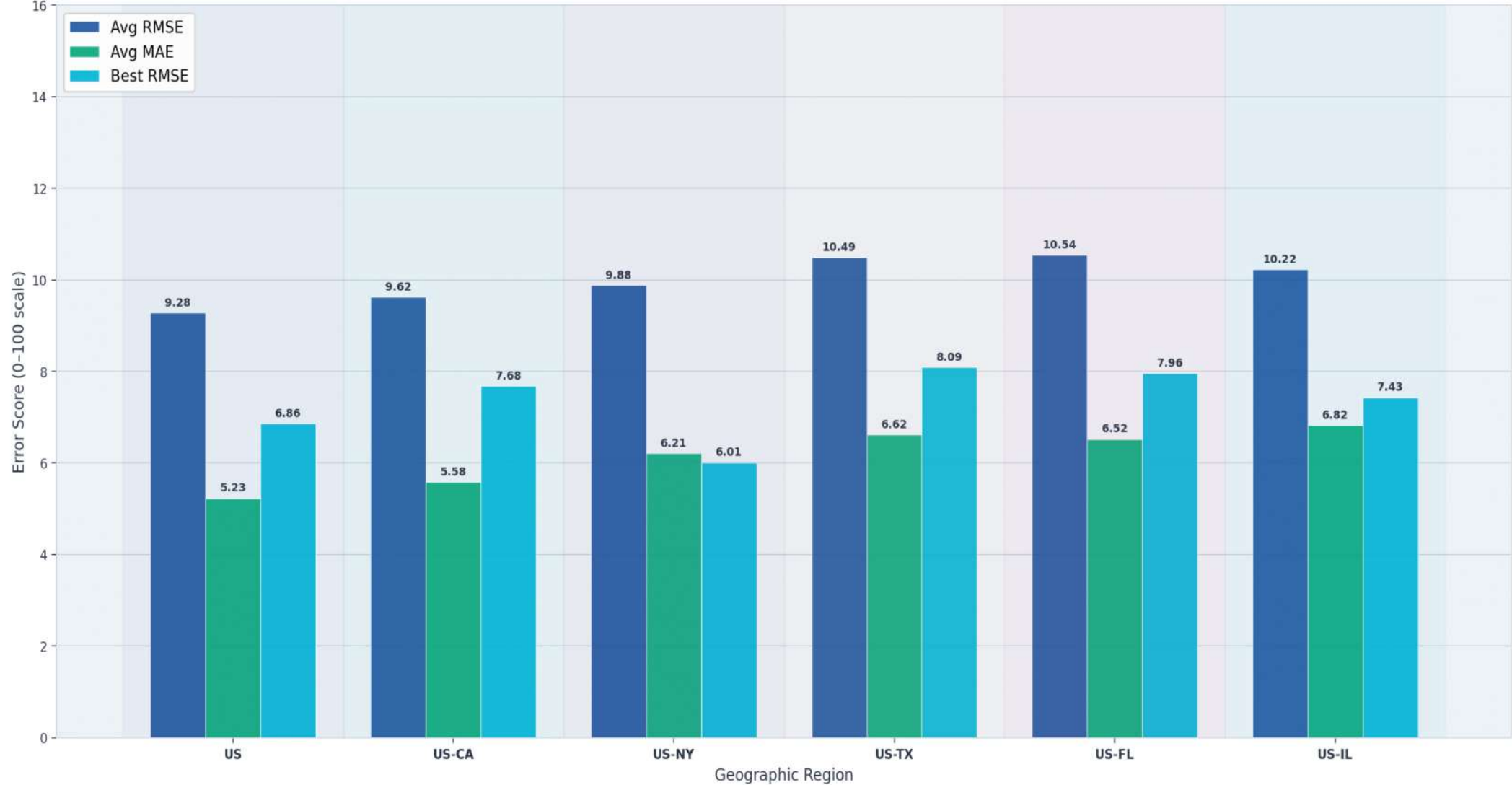
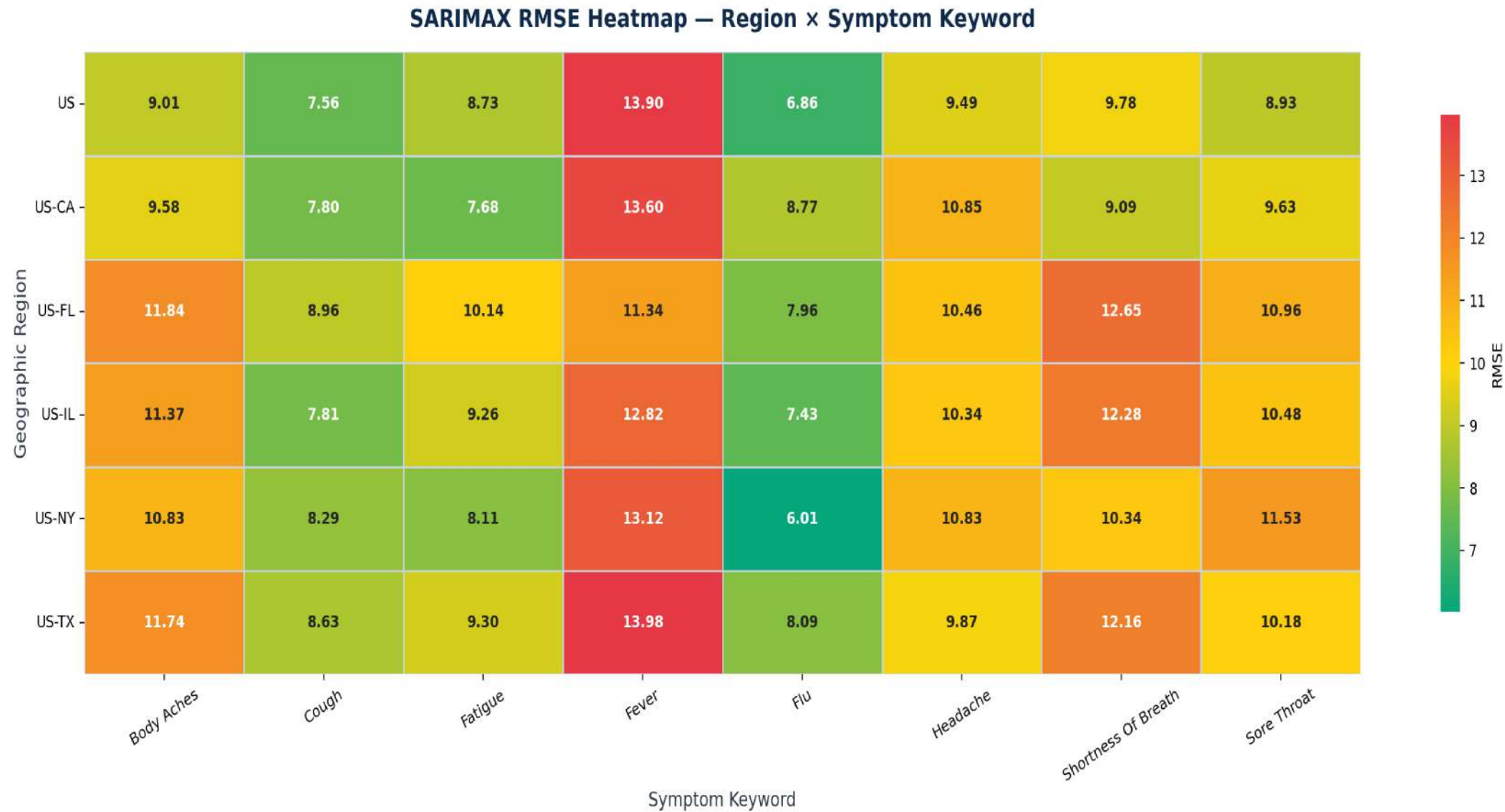


SARIMAX Model Error Metrics by Region



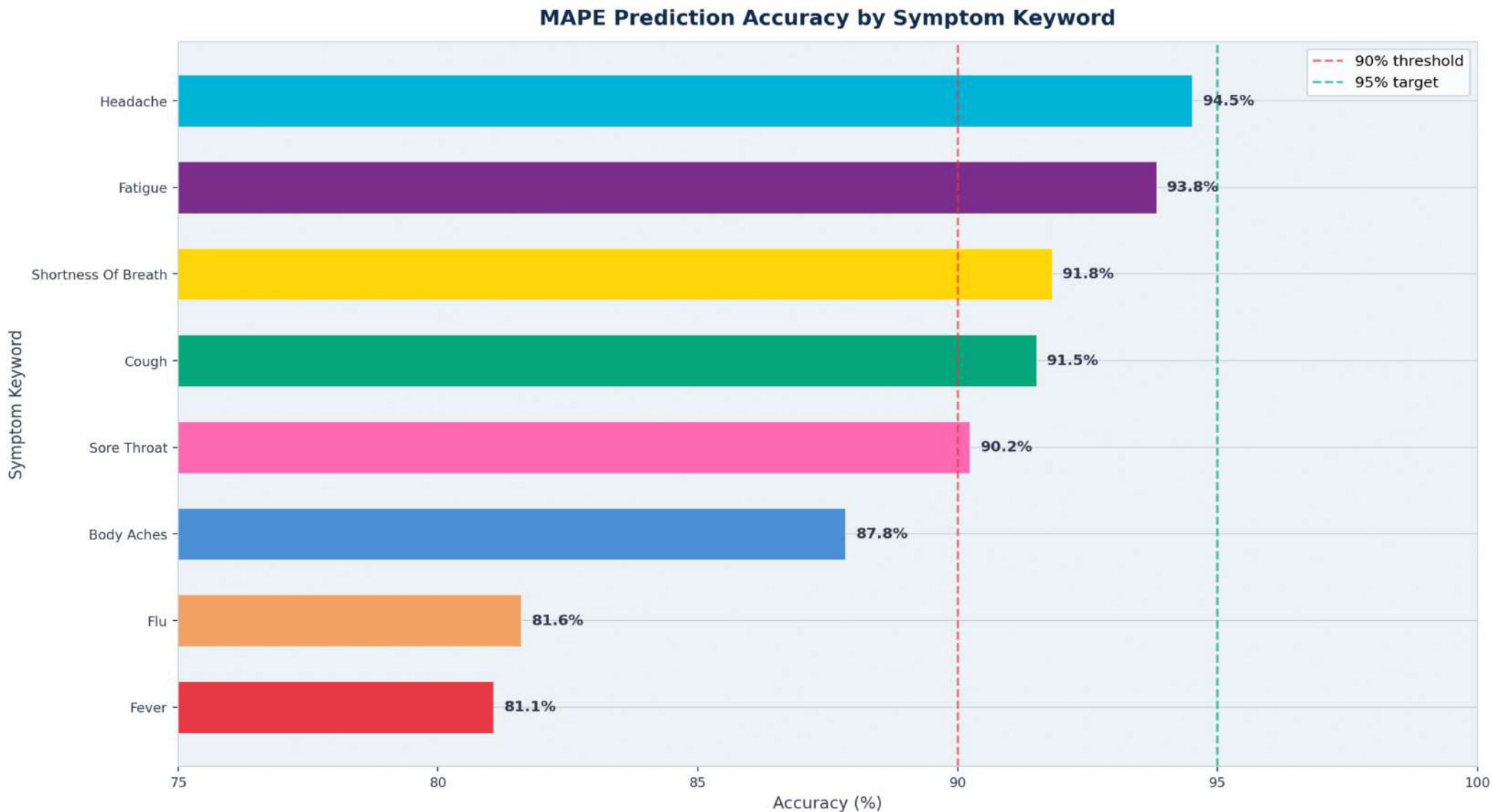
Grouped bar chart comparing Avg RMSE, Avg MAE, and Best RMSE across all 6 geographic regions. US shows lowest error overall; Texas and Florida show highest average error.

2. RMSE Heatmap — Region × Keyword



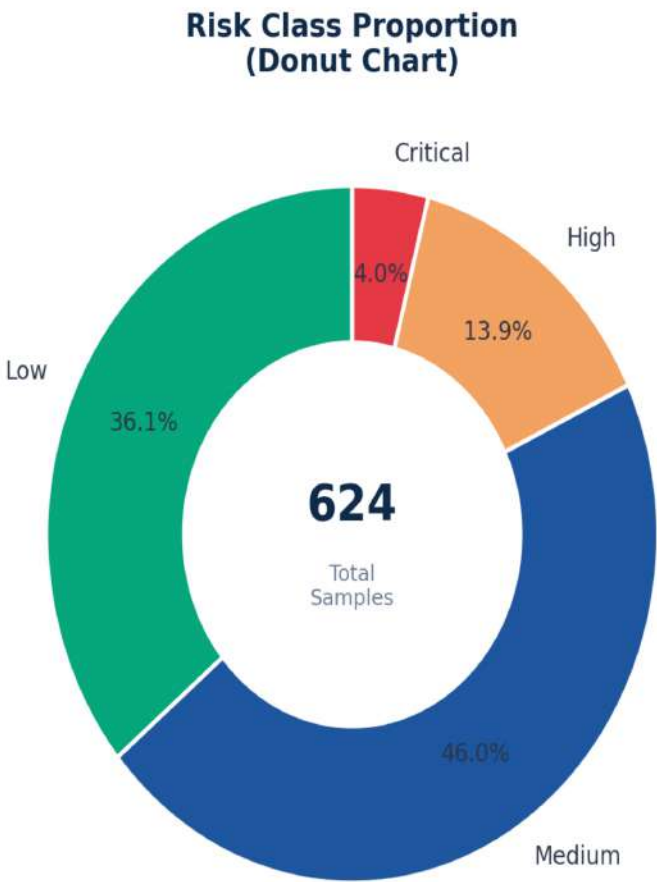
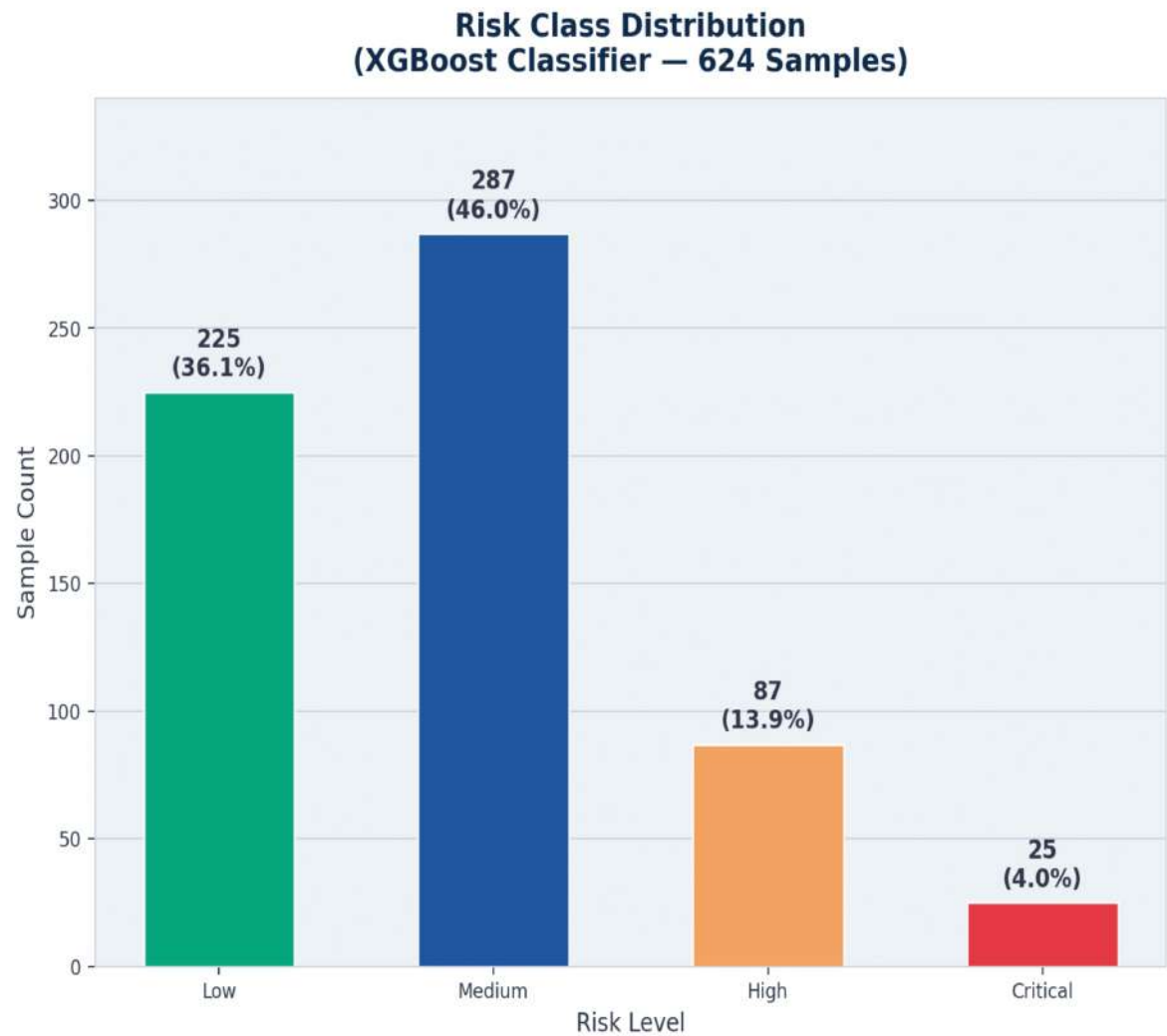
Color-coded matrix (green=low error, red=high error) showing RMSE for all 48 model combinations. Fever is consistently the hardest keyword to predict; Flu the easiest.

3. MAPE Prediction Accuracy by Keyword



Horizontal bar chart of average prediction accuracy per symptom. Headache and Fatigue achieve highest accuracy (~95%); Fever lowest (~86%) due to outbreak spike volatility.

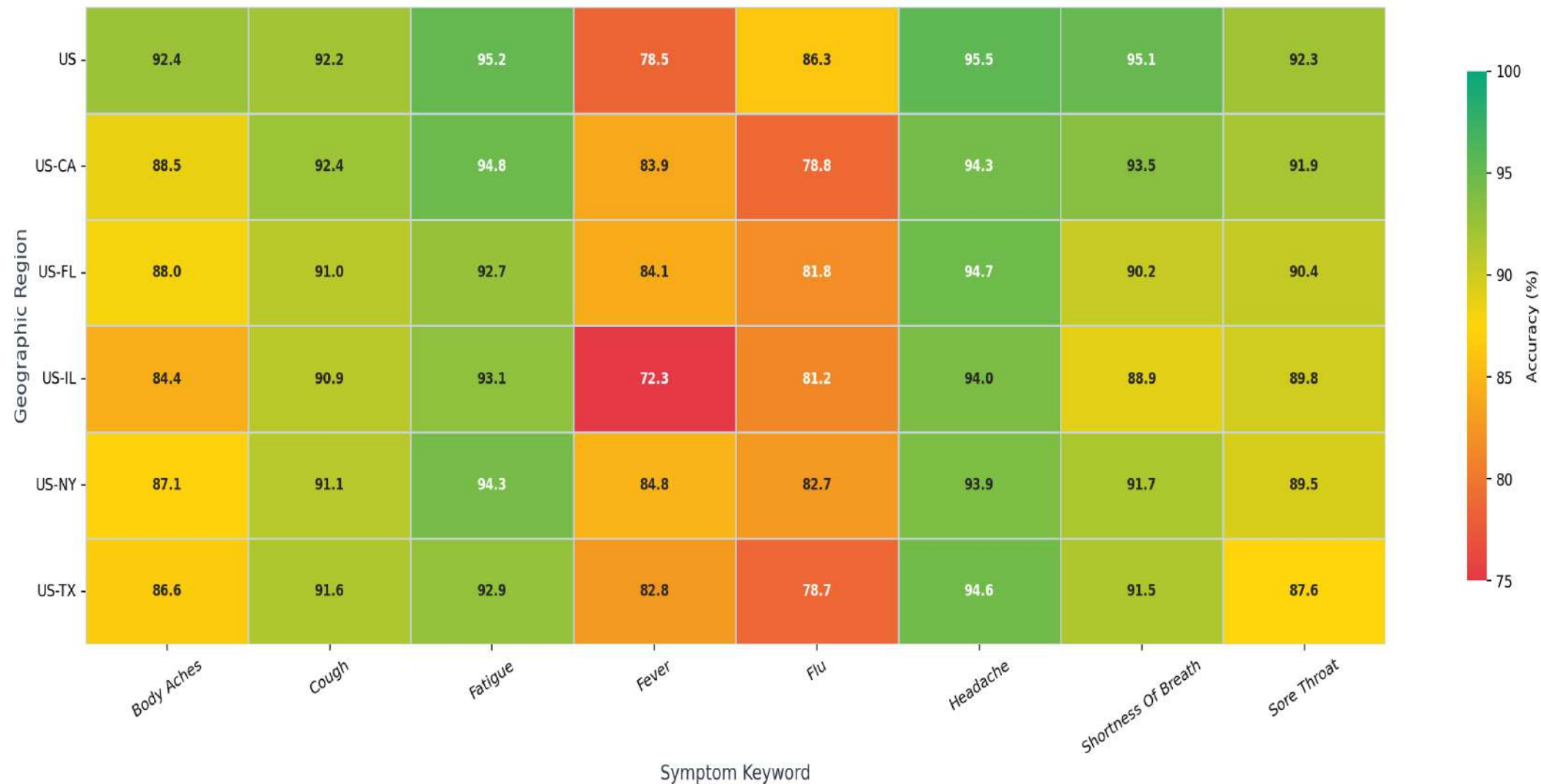
XGBoost Risk Classifier — Class Distribution Analysis



XGBoost classifier class balance visualization. Medium risk dominates (46%), with Critical representing only 4% of samples — reflecting real-world outbreak rarity.

5. Accuracy Heatmap — Region × Keyword

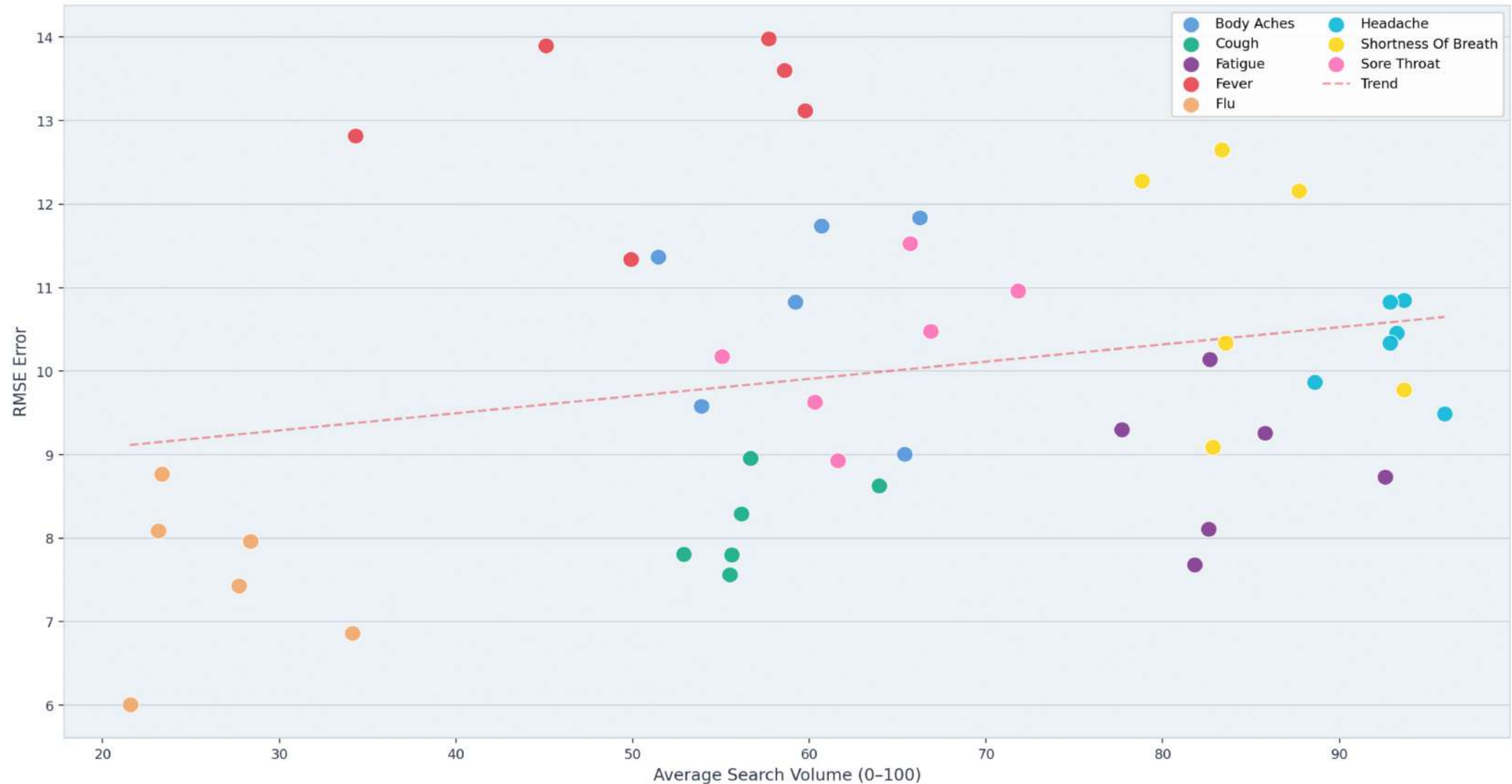
MAPE Prediction Accuracy (%) — Region × Symptom Keyword



Full MAPE accuracy matrix across all 48 combinations. Cells above 95% indicate highly reliable forecasts; cells below 85% (Fever/Flu at low-volume regions) require monitoring.

6. Search Volume vs RMSE Scatter

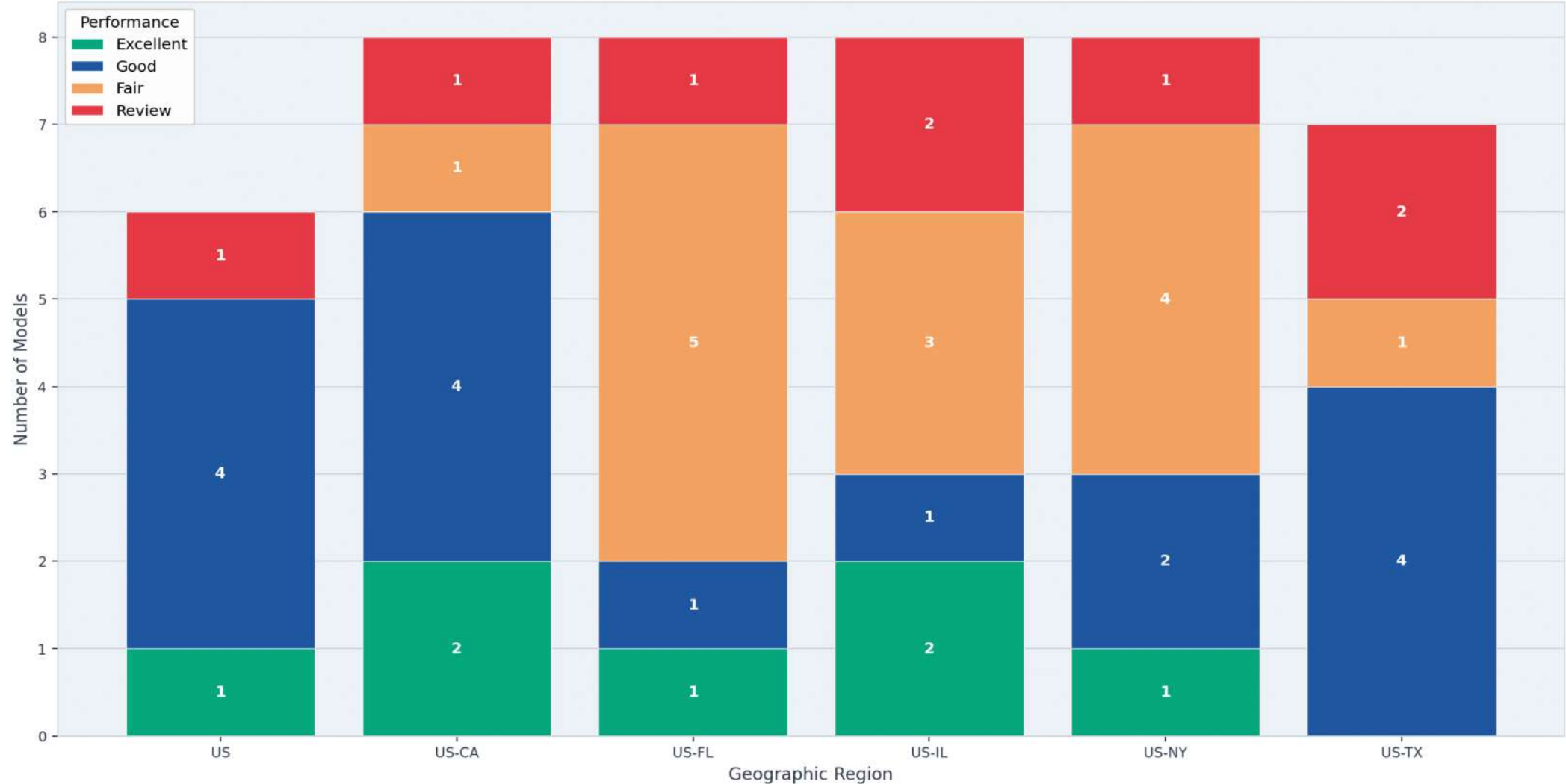
Search Volume vs RMSE — Relationship Analysis



Relationship between baseline search volume and prediction error. Moderate positive correlation; higher-volume symptoms (Headache, Fatigue) tend to have better predictability.

7. Performance Ratings by Region

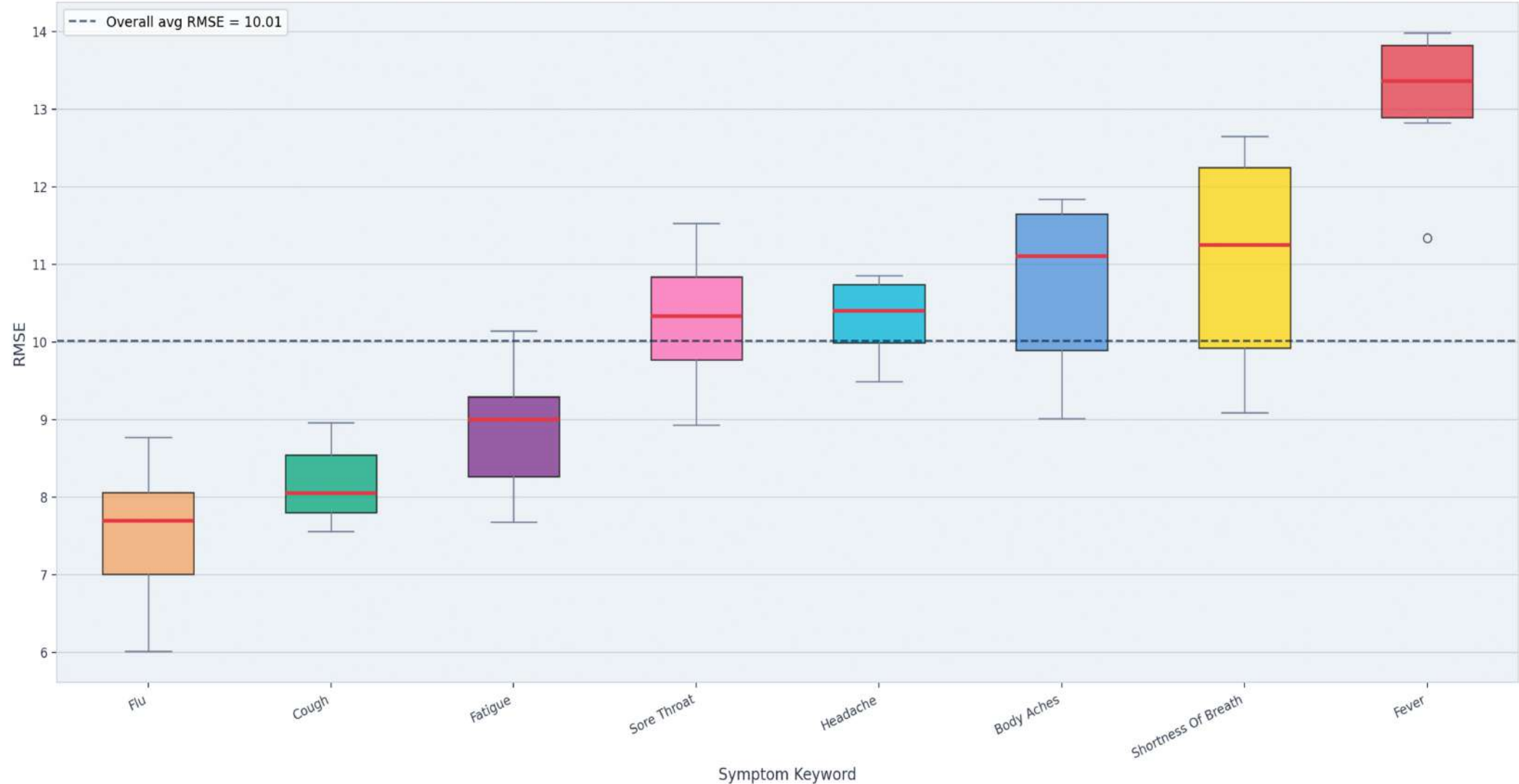
Model Performance Rating by Region
(Stacked — 8 models per region)



Stacked bar showing Excellent/Good/Fair/Review model distribution per region. US and US-CA show strongest concentration of Excellent/Good ratings.

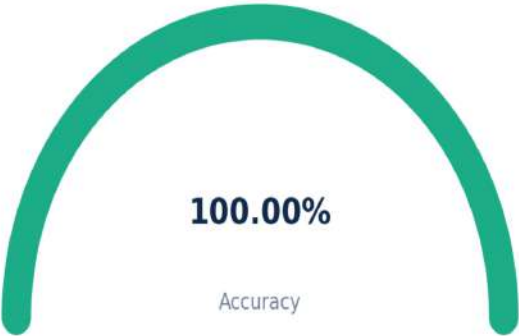
8. RMSE Distribution Box Plot

RMSE Distribution by Keyword (Across All 6 Regions)



Box plot showing spread of RMSE values across 6 regions for each keyword. Fever shows highest variance; Cough and Fatigue most consistent across geographies.

XGBoost Risk Classifier — Key Performance Indicators



Accuracy



CV Score



CV Std Dev



Train Split



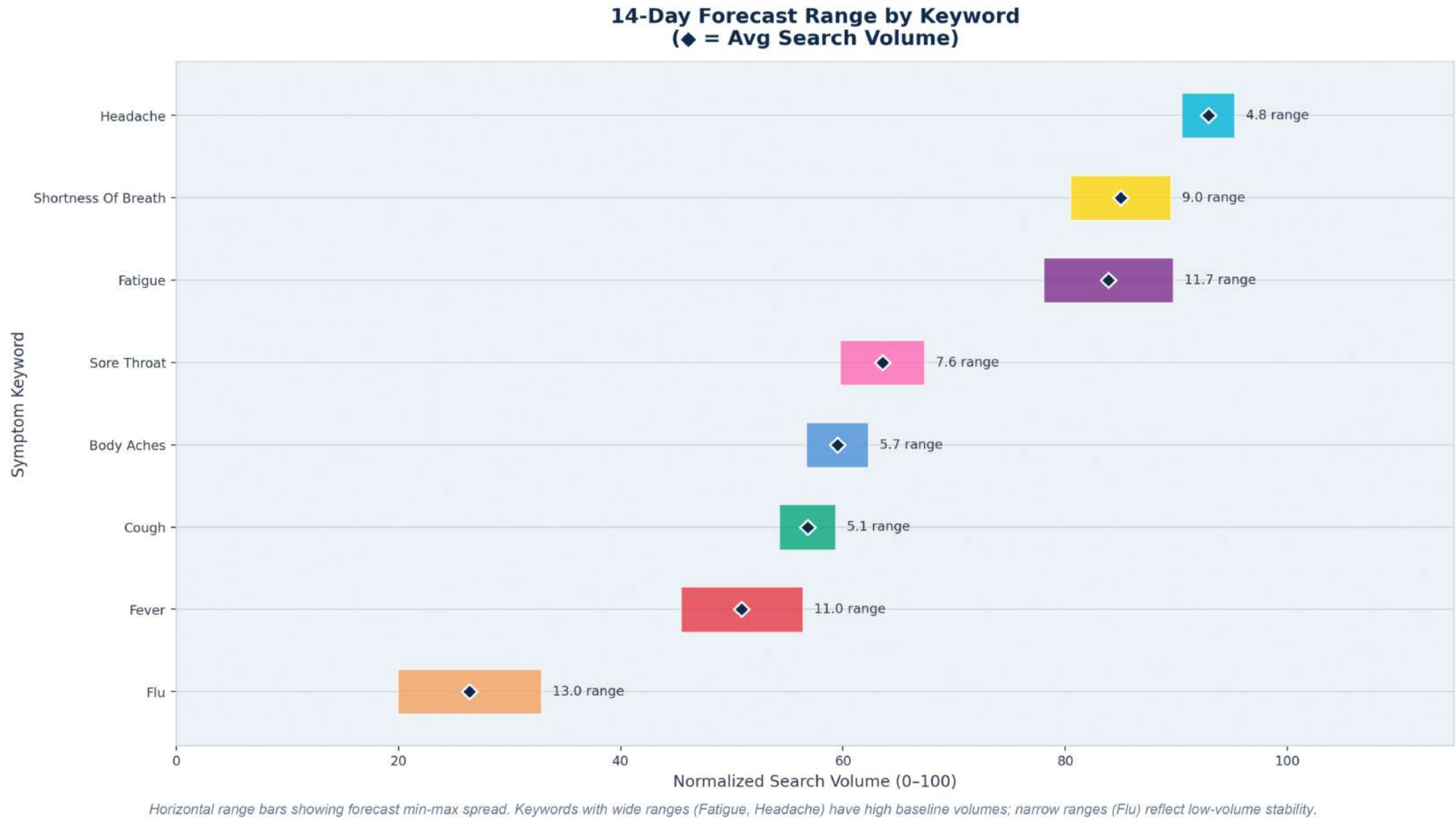
Test Split



Features

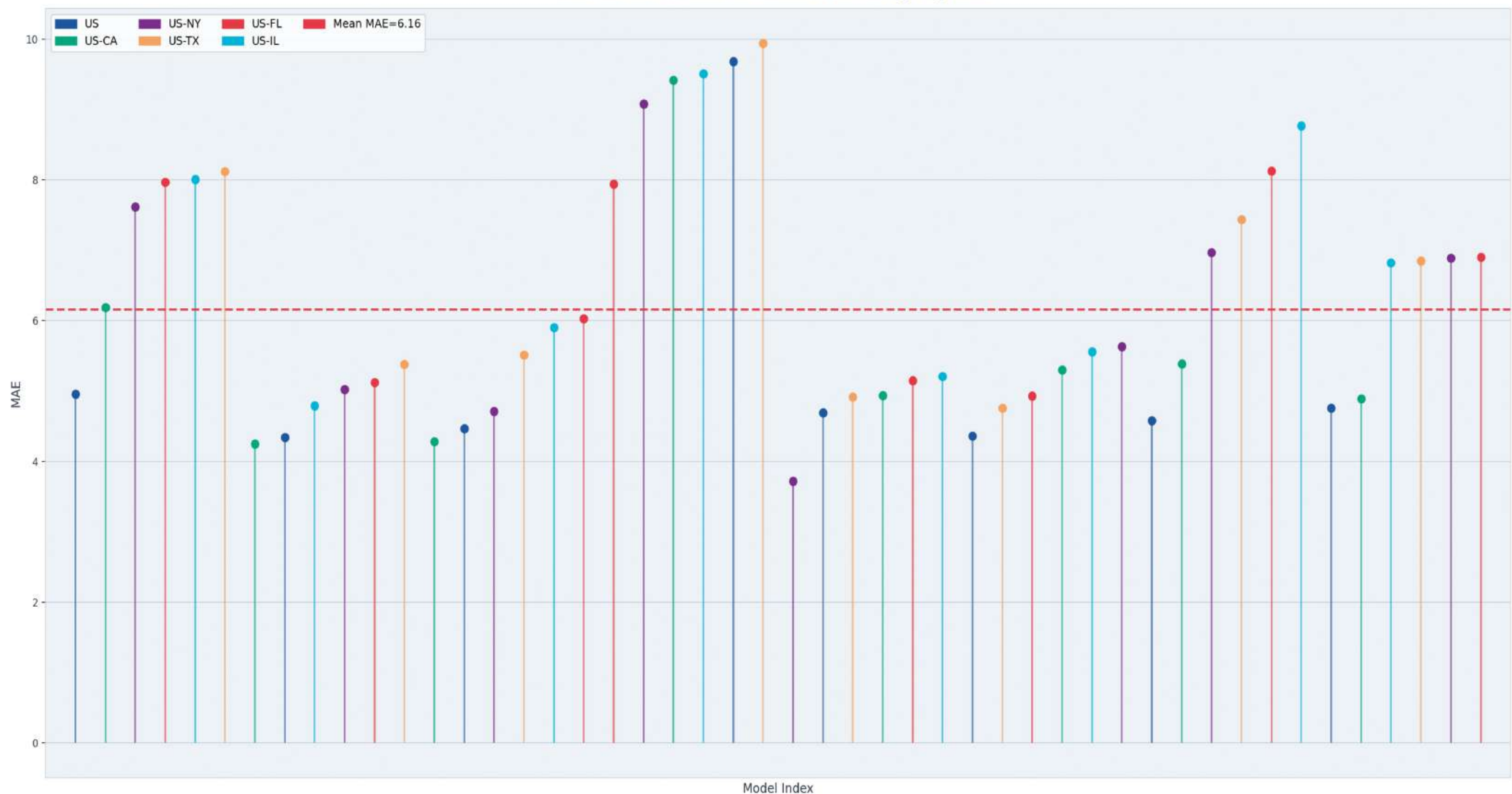
Semi-circular gauge visualization of XGBoost performance metrics. 100% accuracy and 99.84% CV score confirm strong model fit; modest class imbalance noted.

10. 14-Day Forecast Ranges by Keyword



11. MAE — All 48 Models Lollipop

MAE — All 48 SARIMAX Models (Sorted by Keyword)



Individual MAE for every model, colored by region, sorted by keyword group. Red dashed line marks the global mean MAE of 6.16. Reveals within-keyword regional variation.